

Lesson 16 Smallest Counterexample

Prove the following by smallest counterexample.

1. For every natural number n ,

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2$$

2. For every natural number n ,

$$3|(4^n - 1)$$

3. For every natural number n ,

$$n! \leq n^n$$

4. Every integer greater than 1 has a prime factor.

5. Every natural number is even or odd.